

Decision Log

Project

PotholeWatch AI

Purpose

This document records the major design, technical, and product decisions made during the development of the project. It helps explain why the solution was built the way it was and what trade-offs were accepted for a functional MVP.

Decision 1: Build a citizen-first civic reporting workflow

Decision

We designed the app around a simple reporting journey: citizen uploads photo + location + issue details.

Why

- Easy for users to understand and use
- Improves the quality of complaint submission
- Makes the evidence more actionable for civic teams

Impact

This creates a strong user experience and reduces the friction usually associated with civic reporting.

Status

Implemented

Decision 2: Use AI to verify image evidence before accepting a report

Decision

The app uses AI models to decide whether an uploaded image clearly shows a supported civic issue.

Why

- Reduces false or irrelevant submissions
- Helps validate real urban problems
- Improves trust in the reporting system

Impact

The platform becomes more reliable by avoiding blind acceptance of unclear or unrelated images.

Status

Implemented

Decision 3: Add duplicate detection before creating a complaint

Decision

Before finalizing a complaint, the system checks for similar issue types and nearby locations.

Why

- Prevents multiple duplicate reports for the same problem
- Saves civic resources and time
- Makes complaint handling more efficient

Impact

This makes the app more practical for real municipal workflows and reduces noise in the dataset.

Status

Implemented

Decision 4: Validate location data before routing complaints

Decision

The application validates coordinates and checks whether the report falls within the configured Mysuru service area.

Why

- Keeps the app realistic and scoped
- Reduces invalid or misplaced reports
- Improves routing confidence

Impact

It ensures that issue routing is based on usable and relevant geographic information.

Status

Implemented

Decision 5: Keep the solution focused on Mysuru as a pilot city

Decision

We scoped the project to Mysuru instead of building a generic all-city platform.

Why

- Easier to validate the model and workflow
- More realistic for an MVP
- Aligns with the local civic problem context

Impact

This reduces complexity while keeping the product relevant and demonstrable.

Status

Implemented

Decision 6: Use a public dashboard for transparency

Decision

We included a public dashboard to display complaint information without exposing personal data.

Why

- Increases accountability
- Makes citizens aware of civic issue visibility
- Promotes trust in the reporting process

Impact

The system is not only a reporting tool but also a transparency layer for the city and residents.

Status

Implemented

Decision 7: Use Next.js + Supabase for rapid MVP development

Decision

The project was built using Next.js, React, TypeScript, and Supabase.

Why

- Fast development for a hackathon-style product
- Simple UI and API handling
- Easy data persistence and retrieval
- Suitable for prototype deployment

Impact

This architecture enables quick iteration and clear demonstration of the core concept.

Status

Implemented

Decision 8: Keep the project as an MVP, not a full government system

Decision

The project is intentionally designed as a prototype and not as a fully deployed official municipal system.

Why

- Scope is realistic for the timeframe
- Legal and civic integration requirements are broader than the MVP
- Focus remains on demonstrating value and feasibility

Impact

The app clearly communicates the concept while staying practical and credible.

Status

Implemented

Final notes

This project demonstrates how AI, civic data, and a citizen-facing workflow can improve how urban issues are reported, verified, tracked, and resolved. The decisions above were chosen to maximize clarity, usability, and practical value within the constraints of an MVP.